

PAPER_242 — Rings of Relativity: Einstein Ring Lensing Amplification in the Full MUGE

GAL-CLUS-022058s — Static Einstein-Ring Lensing Factor in the Master Universal Gravity Equation

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Framework: Unified Quantum Field Framework (UQFF) v4.10

Calculator Class: RingsOfRelativityEinsteinLensingMUGECalculator (CP3, Session 60)

Encoded By: Grok (xAI), October 2025 (C++ source); Python CP3 integration March 2026

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Abstract

This paper presents the full Master Universal Gravity Equation (MUGE) for the GAL-CLUS-022058s gravitational lens system (“Rings of Relativity”). The key unique mathematical contribution is the **static Einstein ring lensing amplification factor** L_t , derived from the angular diameter distance ratio D_{LS}/D_S and the Schwarzschild radius of the galaxy cluster lens, applied as a multiplicative correction to the base gravitational field.

This is expressly **distinct** from the dynamic lensing modulation $L(t) = L_0 e^{-t/\tau} \cos(\omega t)$ already captured in CP3 class 81 (UQFFLensingModulationRingsCalculator, Session 53). The present paper captures the static, geometry-driven Einstein ring correction.

2. System Parameters

Parameter	Symbol	Value	Units
Lens mass	M	$10^{14} M_{\odot}$	kg
Einstein radius	r	3.086×10^{20}	m (~10 kpc)
Redshift of lens	z_{lens}	0.5	dimensionless
Angular distance ratio	$L_{\text{factor}} = D_{LS}/D_S$	0.67	dimensionless
Magnetic field	B	10^{-5}	T
Gas velocity	v_{gas}	10^5	m/s
Oscillatory amplitude	A_{osc}	10^{-12}	m/s ²

3. The Novel Lensing Amplification Term

3.1 Einstein Ring Lensing Factor

The static gravitational lensing amplification of the measured gravitational field along the line of sight through the Einstein ring geometry:

$$L_t = \frac{GM}{c^2 r} \cdot L_{\text{factor}}, \quad L_{\text{factor}} = \frac{D_{LS}}{D_S} = 0.67$$

$$\text{corr}_L = 1 + L_t$$

where: - GM/c^2r is the dimensionless Schwarzschild-radius-to-Einstein-radius ratio - D_{LS} = angular diameter distance from lens to source - D_S = angular diameter distance from observer to source - The factor D_{LS}/D_S encodes the lensing geometry in the single-lens single-source approximation

3.2 Friedmann Correction at $z = 0.5$

$$H(z = 0.5) = H_0 \sqrt{\Omega_m(1+z)^3 + \Omega_\Lambda} = H_0 \sqrt{0.3 \cdot (1.5)^3 + 0.7}$$

This mid-redshift Friedmann correction is distinct from $z = 0$ (local), $z = 3.5$ (HUDF), or $z = 0.01$ (nearby lenses) used in earlier classes.

4. Full MUGE: All Nine Terms

$$g_{\text{Rings}}(r, t) = \underbrace{T_1}_{\text{base+H+B+L}} + \underbrace{T_2}_{\text{UQFF}} + \underbrace{T_3}_{\Lambda} + \underbrace{T_4}_{\text{EM}} + \underbrace{T_q}_{\text{QM}} + \underbrace{T_{\text{fl}}}_{\text{fluid}} + \underbrace{T_{\text{osc}}}_{\text{2-mode}} + \underbrace{T_{\text{DM}}}_{\text{DM+pert2}} + \underbrace{T_{\text{wind}}}_{\text{stellar wind}}$$

Term 1 — Base gravity with $H(z)$, magnetic, Einstein lensing:

$$T_1 = \frac{GM}{r^2} (1 + H(z)t) (1 - B/B_{\text{crit}}) (1 + L_t)$$

Term 2 — UQFF unified field with time-reversal:

$$T_2 = (U_{g1} + U_{g4})(1 + f_{TRZ})$$

Term 3 — Cosmological constant:

$$T_3 = \frac{\Lambda c^2}{3}$$

Term 4 — Electromagnetic with vacuum density ratio:

$$T_4 = \frac{qv_{\text{gas}}B}{m_p} \left(1 + \frac{\rho_{\text{UA}}}{\rho_{\text{SCm}}} \right) \cdot s_{\text{EM}}, \quad \frac{\rho_{\text{UA}}}{\rho_{\text{SCm}}} = \frac{7.09 \times 10^{-36}}{7.09 \times 10^{-37}} = 10$$

Term 5 — Quantum uncertainty:

$$T_q = \frac{\hbar}{\sqrt{\Delta x \Delta p}} \psi \frac{2\pi}{t_H}$$

Term 6 — Fluid (buoyancy-acceleration):

$$T_{\text{fl}} = \frac{\rho_{\text{fluid}} V g_{\text{base}}}{M}$$

Term 7 — Two-mode oscillation (standing wave + traveling wave):

$$T_{\text{osc}} = 2A \cos(kx) \cos(\omega t) + \frac{2\pi}{t_{\text{Gyr}}} A \cos(kx - \omega t)$$

The first mode is a **standing wave decomposition** ($2 \cos \cos$); the second mode is a **Gyr-scaled traveling wave**. Together they form a beating-mode pair unique to this system.

Term 8 – Dark matter with $\delta_2 = 3GM/r^3$ perturbation:

$$T_{\text{DM}} = \frac{(M + M_{\text{DM}})(\delta\rho/\rho + 3GM/r^3)}{M}$$

The second-order perturbation $3GM/r^3$ is a tidal-force density correction distinct from density-contrast $\delta\rho/\rho$.

Term 9 – Stellar wind feedback:

$$T_{\text{wind}} = \frac{\rho_{\text{wind}} v_{\text{wind}}^2}{\rho_{\text{fluid}}}$$

5. Distinction from Prior Art

Feature	Class 81 (Session 53)	This class (Session 60)
Lensing factor	$L(t) = L_0 e^{-t/\tau} \cos(\omega_{\text{lens}} t)$	$L_t = (GM/c^2 r) \cdot (D_{LS}/D_S)$
Physics	Dynamic lens transit alignment	Static Einstein ring geometry
Time dependence	Yes (decaying oscillation)	No (geometric constant)
Parameter	$L_0 = 0.15, \tau_{\text{lens}}, \omega_{\text{lens}}$	$L_{\text{factor}} = 0.67$ (distance ratio)
Oscillation	None	Two-mode standing + traveling wave
DM pert2	Not present	$3GM/r^3$ tidal correction
Wind term	Not present	$\rho_w v_w^2 / \rho_{\text{fl}}$

6. Numerical Verification

Using default parameters ($M = 1.989 \times 10^{44}$ kg, $r = 3.086 \times 10^{20}$ m, $z = 0.5$):

$$L_t = \frac{(6.6743 \times 10^{-11})(1.989 \times 10^{44})}{(2.998 \times 10^8)^2 (3.086 \times 10^{20})} \times 0.67 \approx 1.6 \times 10^{-3}$$

$$\text{corr}_L \approx 1.0016 \quad (\sim 0.16\% \text{ amplification})$$

$$H(z = 0.5) = H_0 \sqrt{0.3 \times 3.375 + 0.7} \approx 1.27 H_0$$

The lensing correction is small but non-zero and physically motivated by the Einstein ring geometry of GAL-CLUS-022058s ($\theta_E \approx 10''$, confirmed by HST imaging).

7. Integration Into UQFF Pipeline

- **CP3 class:** RingsOfRelativityEinsteinLensingMUGECalculator (112th calculator, Session 60)
 - **Aggregator:** v2.4.0 — registered in CP3_CALCULATORS
 - **Source:** Doc 8 C++ class RingsOfRelativity (Grok/xAI, October 2025)
 - **Complementary classes:**
 - Class 81 UQFFLensingModulationRingsCalculator — dynamic lens transit
 - Class 111 NGC3603FullMUGECavityPressureCalculator — companion new class (PAPER_243)
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8. References

- Hubble Space Telescope imaging of GAL-CLUS-022058s (Infante et al. 2022)
 - Einstein (1936): Lens-star problem, Science 84, 506
 - Schneider, Ehlers & Falco (1992): Gravitational Lenses, Springer
 - Murphy, D.T. (2025): UQFF Manuscript — Doc 8 (Rings of Relativity MUGE, October 2025)
 - Grok (xAI): C++ class RingsOfRelativity, encoded October 2025
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